Statistics
The Complete Visual Tutorial

From your very first frequency table to logistic regression and K-means clustering — every concept explained from scratch, with the math laid out plainly and a picture next to every idea.

Descriptive Statistics | Hypothesis Testing | T-Tests & ANOVA | Correlation & Regression | Clustering | Confidence Intervals

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What Is Statistics?

Statistics deals with the **collection, analysis, and presentation of data**. That's the whole job in one sentence — everything else in this tutorial is really just different tools for doing those three things well.

EXAMPLE

Suppose you want to know whether **gender** influences someone's **preferred newspaper**. Gender and newspaper are your *variables*. To study them, you write a questionnaire, send it out, and after a couple of weeks you get back a table: one column per variable, one row per person. Row 1 might read "male, New York Post"; row 2 "female, USA Today." That table is your raw data — it could just as easily have come from an experiment, e.g. testing two drugs on blood pressure instead of a survey.

Once you have that table, the real question begins: are you happy just describing the people you actually surveyed (your **sample**), or do you want to say something about everyone the sample represents (the **population**)? That single question splits all of statistics into two branches.

Descriptive vs. Inferential Statistics

DESCRIPTIVE STATISTICS

Summarizes and describes the data you actually collected — nothing more. If you know how 200 employees at one company commute to work, descriptive statistics lets you say "58% drive." It says nothing about employees anywhere else.

INFERENTIAL STATISTICS

Uses the sample to draw a conclusion about the wider population. It lets you take those same 200 employees and make an educated, quantified guess about how *all* employees — including ones you never surveyed — behave.

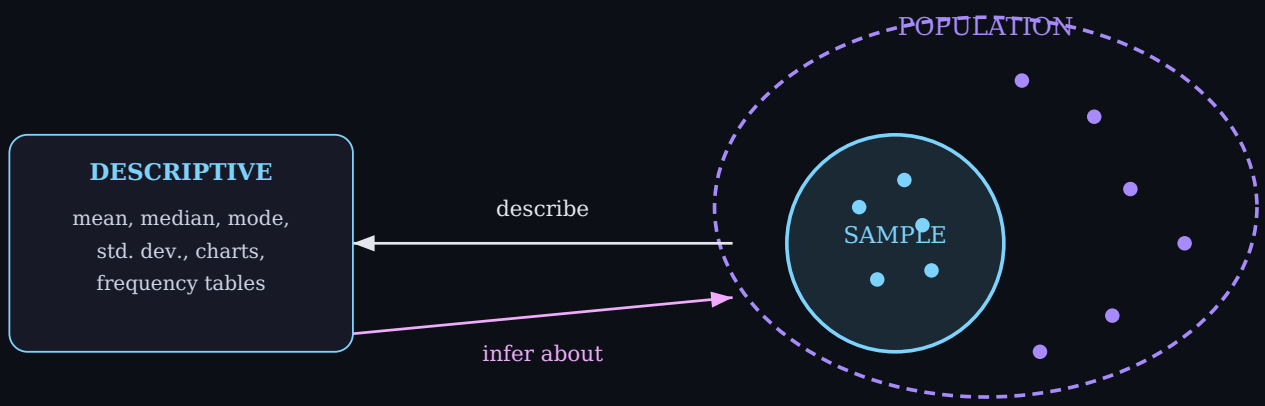


Fig. 1 — You always start with a sample drawn from a population. Descriptive statistics stays inside the sample; inferential statistics reaches out to make a claim about the population.

To describe a data set, we lean on four building blocks, covered in Part 2: **measures of central tendency**, **measures of dispersion**, **frequency tables**, and **charts**. To make claims about a population, we lean on **hypothesis testing**, covered starting in Part 3.

Levels of Measurement

Before you can analyze a variable correctly, you need to know what *kind* of variable it is — its **level of measurement**. This single decision determines which averages make sense, which charts to draw, and which hypothesis tests are even legal to run. Get it wrong at the start of a project and no amount of clever analysis at the end will save you.

Level	Can you…	Examples	Typical chart
Nominal	…categorize, but NOT rank	gender, blood type, preferred newspaper, telephone numbers (numeric but nominal!)	Bar chart / frequency table
Ordinal	…categorize AND rank, but gaps aren't equal	satisfaction rating, education level, care level (low/medium/high)	Ordered bar chart
Interval	…rank with equal gaps, but no true zero	temperature in °C, a stopwatch that started late	Histogram
Ratio	…rank, equal gaps, AND a true zero (ratios are meaningful)	age, income, weight, CO_2 emissions, marathon time from the gun	Histogram

SHORTCUT

Because interval and ratio data are analyzed almost identically, they're often lumped together as **metric data**. So in practice you mostly need to decide between three buckets: **nominal**, **ordinal**, or **metric**.

EXAMPLE — THE MARATHON

If every runner's watch starts at zero when the gun fires, finishing times are **ratio** data: a 2-hour finisher really was 3× faster than a 6-hour finisher, because "0" means "no time has passed." But if the timer was forgotten and only the *gaps between runners* were recorded, there's no true zero anymore — you can say Runner B finished "1 hour after" Runner A, but not that Runner A was "3× faster." That's **interval** data: equal spacing, no meaningful ratios.

WHY THIS MATTERS

The mean and standard deviation are appropriate for metric data. They can sometimes be defended for ordinal data if you're careful about interpretation, but they make **no sense** for nominal data — you cannot average "car, bicycle, walk." The level of measurement also decides which hypothesis test is even legal to run later on, so nail it down before you collect a single row of data, not after.

Measures of Central Tendency

A measure of central tendency is a single number that tries to represent an entire data set. There are three: the **mean**, the **median**, and the **mode**.

The Mean

FORMULA — ARITHMETIC MEAN

$$\bar{x} = \frac{\Sigma x_i}{n} = \text{sum of all observations} \div \text{number of observations}$$

EXAMPLE

Five students score 78, 85, 90, 88, 92 on a test. Mean = (78+85+90+88+92)/5 = **86.6**.

The Median

Sort the data. If **n is odd**, the median is the middle value. If **n is even**, it's the average of the two middle values.

KEY PROPERTY

The median is **robust to outliers** — the person in the middle stays in the middle no matter how extreme the last value gets. The mean has no such protection: one huge outlier drags it around.

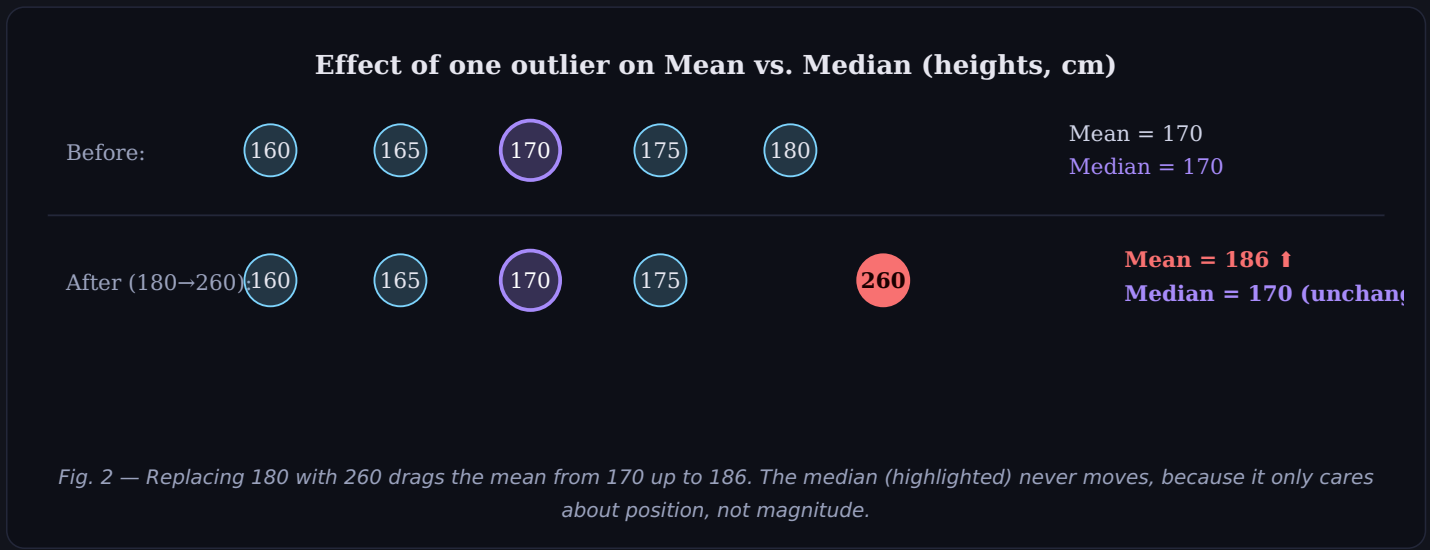


Fig. 2 — Replacing 180 with 260 drags the mean from 170 up to 186. The median (highlighted) never moves, because it only cares about position, not magnitude.

The Mode

The value that occurs most often. It's the only measure of central tendency that works for **nominal** data.

EXAMPLE

Commute survey: 14 car, 6 bike, 5 walk, 5 public transport → the mode is **car**.

Measures of Dispersion

Central tendency tells you where the data is *centered*. It says nothing about how *spread out* it is — and two data sets with

an identical mean can look completely different. That's what measures of dispersion capture: **variance**, **standard deviation**, **range**, and the **interquartile range (IQR)**.

Standard Deviation & Variance

The standard deviation is the average distance between each data point and the mean.

FORMULA — STANDARD DEVIATION

$$\sigma = \sqrt{\left(\frac{\Sigma(x_i - \bar{x})^2}{n}\right)}$$

N VS. N − 1

There are two versions of this formula. Divide by **n** when you truly have the whole population. Divide by **n − 1** whenever your data is a *sample* meant to estimate a population value (e.g. a survey or a clinical study) — which, in practice, is almost always.

Variance is simply the standard deviation squared (σ^2) — it's the same idea, just on a squared scale, which is why standard deviation (same units as the original data) is usually the one people report and interpret.

Range & Interquartile Range (IQR)

- **Range** = maximum − minimum. Simple, but very sensitive to a single outlier.
- **IQR** = Q3 − Q1, the span of the *middle 50%* of the data (25% of values fall below it, 25% above). Much more robust, because extreme outliers can't stretch it.

CENTRAL TENDENCY VS. DISPERSION, SIDE BY SIDE

Imagine measuring blood pressure in two patient groups with the exact same mean. Central tendency (mean, median, mode) tells you where the "typical" value sits. Dispersion (SD, range, IQR) tells you whether patients are tightly clustered around that typical value or scattered widely around it — and that difference can be the whole story.

Frequency Tables, Contingency Tables & Charts

Frequency Tables

A frequency table shows how often each distinct value of **one** variable appears. For the 30-employee commute survey, it would list each transport option and its count.

Mode of transport	Count	%
Car	14	47%
Bicycle	6	20%
Walk	5	17%
Public transport	5	17%

Contingency Tables (Cross-Tabs)

What if you have **two** categorical variables — say, transport mode *and* which factory site (Detroit vs. Cleveland) an employee works at? A contingency table puts one variable in the rows and the other in the columns; each cell counts how many people fall into that exact combination.

	Car	Bicycle	Walk	Public
Detroit	6	3	2	3
Cleveland	8	3	3	2

Reading the table: "car" and "Detroit" were answered together 6 times.

Choosing the Right Chart

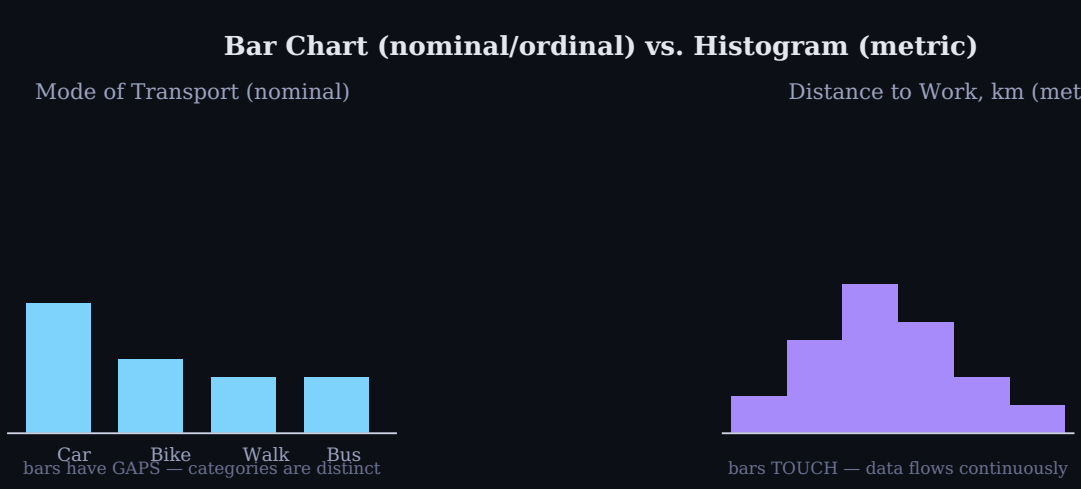


Fig. 3 — Bar charts (with gaps) suit nominal/ordinal categories. Histograms (touching bars) suit continuous metric data, and their shape is exactly what you'll use later to judge normality.

Level of measurement	Best chart(s)
Nominal	Bar chart, pie chart, frequency table
Ordinal	Ordered bar chart
Metric (interval/ratio)	Histogram, box plot, violin plot

One metric variable across groups	Grouped bar chart of means, box plot per group

Hypothesis Testing, From Scratch

This is the conceptual heart of the whole tutorial. Every test in Part 4 — t-tests, ANOVA, chi-square, everything — is just this same six-step logic applied to a different situation.

The Six Steps

1. **Hypothesis.** State a claim you want to test — e.g. "this drug lowers blood pressure."
2. **Population & sample.** You can't measure everyone with high blood pressure, so you take a sample.
3. **Null vs. alternative hypothesis.** You can't directly test "the drug works," so you test its logical opposite: **H_0 (null): the drug has no effect. H_1 (alternative): the drug has an effect.**
4. **Assume H_0 is true** and ask: if there really is no effect in the population, how likely is it that a random sample would show a difference this large, or larger, just by chance?
5. **That probability is the p-value.**
6. **Decide.** If the p-value is small enough, you reject H_0 in favor of H_1.

> **THE P-VALUE, PRECISELY**
> The p-value is the probability of observing a sample that deviates from H_0 by *as much as*, or more than, the one you actually collected — **assuming H_0 is true**. It is *not* "the probability H_0 is true," a very common misreading.

Statistical Significance

You need a cutoff to decide "small enough." That cutoff is the **significance level (α)**, almost always set at **0.05 (5%)**.

- **$p < 0.05$** → reject H_0 → result is "statistically significant" → the data is inconsistent enough with "no effect" that you conclude there probably is one.
- **$p \geq 0.05$** → fail to reject H_0 → not enough evidence for an effect (this is *not* proof H_0 is true — absence of evidence isn't evidence of absence).

Type I and Type II Errors

Any test can go wrong in two distinct ways:

	H_0 is actually TRUE	H_0 is actually FALSE
You reject H_0	Type I error (false positive)	Correct ✓
You fail to reject H_0	Correct ✓	Type II error (false negative)

> **EXAMPLE**
> **Type I:** the drug does nothing, but your sample happened to look impressive by chance — you wrongly conclude it works.
> **Type II:** the drug genuinely works, but your particular sample happened to show little difference — you wrongly conclude it doesn't.

Directed vs. Undirected (One-Tailed vs. Two-Tailed) Tests

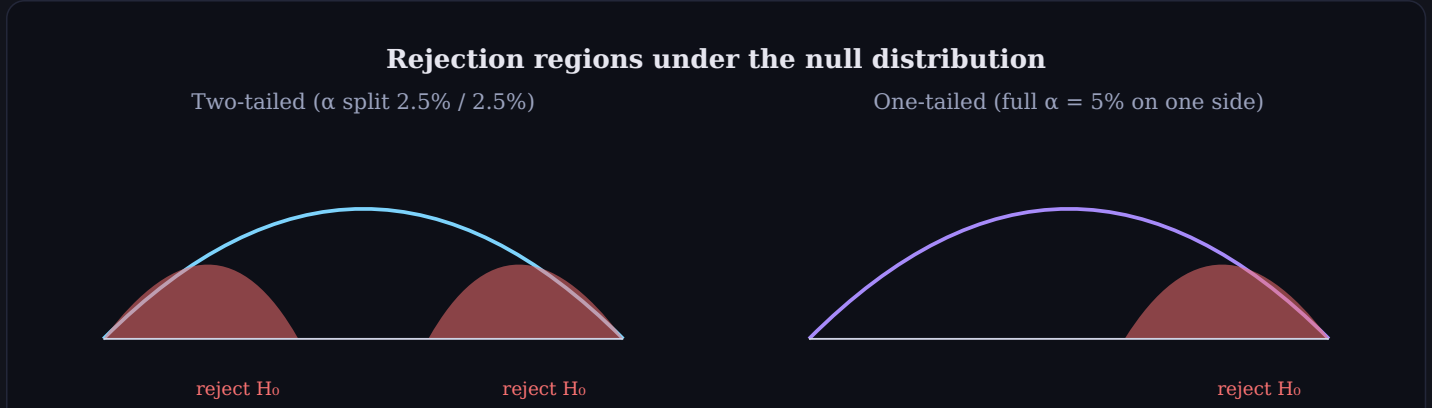


Fig. 4 — Undirected hypotheses ("there IS a difference") split the 5% risk across both tails. Directed hypotheses ("group A is HIGHER than group B") put the full 5% in one tail, making it easier to detect an effect in that specific direction.

Parametric vs. Non-Parametric Tests

Before running any hypothesis test, you must check its **assumptions** — and the biggest one is usually: is the data normally distributed?

	Parametric	Non-parametric
Requires normal distribution?	Yes	No
Works on	Raw values	Ranks of the values
Statistical power	Higher — detects real effects with a smaller sample/difference	Lower, but safer when normality fails
Examples	t-test, ANOVA, Pearson correlation	Mann-Whitney U, Kruskal-Wallis, Spearman correlation

RULE OF THUMB

If possible, always prefer the parametric test — it's more likely to detect a real effect. Only drop to the non-parametric counterpart when an assumption (usually normality) is actually violated.

Testing for Normal Distribution

Two families of methods exist:

ANALYTICAL TESTS

Shapiro-Wilk, Kolmogorov-Smirnov, Anderson-Darling. All test H_0: "the data is normally distributed," and all output a p-value. $p < 0.05 \rightarrow$ not normal.

Catch: the p-value is very sensitive to sample size. Small samples almost always look "normal" (underpowered to detect deviation); huge samples can flag trivial, irrelevant deviations as "significant." This is why graphical checks matter too.

GRAPHICAL TESTS

Histogram overlaid with a normal curve, or — better — a **Q-Q plot**, which plots your data's quantiles against the quantiles a perfect normal distribution would have. Points hugging the diagonal line = normal. Points forming a curve or straying outside the 95% confidence band = not normal.



Fig. 5 — Left: the classic bell shape. Right: in a Q-Q plot, data points hugging the diagonal (dashed) confirm normality — no bell-curve-eyeballing required.

Levene's Test for Equal Variances

Several tests (independent t-test, ANOVA) also assume the groups being compared have **equal variance**. Levene's test checks exactly that.

HYPOTHESES

H_0: the variances of the groups are equal.

H_1: at least one group has a different variance.

If Levene's p-value > 0.05, equal variance can be assumed and you proceed with the standard test. If $p < 0.05$, use the adjusted version of the test built for unequal variances (e.g. Welch's t-test).

The T-Test

The t-test checks whether there's a **significant difference between the means of two groups**. Its dependent variable must be **metric** and (for the test to be valid) **normally distributed**. There are three variants, and choosing between them depends entirely on your study design.

Variant	Use when...	Example
One-sample	Comparing one sample's mean to a known reference value	Do 30 chocolate bars average 50 g as claimed?
Independent-samples	Comparing two separate, unrelated groups	Painkiller A vs. Painkiller B, different people in each group
Paired-samples	Comparing the same people measured twice	Weight before vs. after a diet, same 30 people

NEAT TRICK

The paired-samples t-test is secretly just a one-sample t-test: compute each person's before/after *difference*, and test whether the mean of those differences is significantly different from a reference value of zero.

Hypotheses

Variant	H_0	H_1
One-sample	sample mean = reference value	sample mean ≠ reference value
Independent	mean(group 1) = mean(group 2)	mean(group 1) ≠ mean(group 2)
Paired	mean of the differences = 0	mean of the differences ≠ 0

Assumptions

- The variable being compared is **metric** (e.g. weight, income — not education level).
- It's **normally distributed** in all three variants.
- For the independent-samples version only: variances in the two groups are roughly equal (check with **Levene's test**).

How It's Calculated

THE GENERAL IDEA

$$t = \frac{\text{difference between means}}{\text{standard error (spread of the mean)}}$$

A bigger mean difference → bigger t-value. More scatter in the data (a larger standard error) → smaller t-value, because a given difference means less when the data is noisy. Degrees of freedom: **$n - 1$** for one-sample and paired tests; **$n_1 + n_2 - 2$** for the independent-samples test.

WORKED EXAMPLE — ONE-SAMPLE T-TEST

Claim: chocolate bars weigh 50 g on average. Sample of 30 bars: mean = 48 g. We compute a t-value from the difference (48 − 50) and the standard error, then compare it against the **critical t-value** for 29 degrees of freedom at α = 0.05 — or, more directly, just read off the p-value. Suppose we get **t = 2.5** with 9 degrees of freedom (a smaller illustrative sample): the critical t-value at α=0.05 (two-tailed) is **2.262**. Since 2.5 > 2.262, we reject H_0 —

the bars are significantly lighter than claimed. Equivalently, this t and these degrees of freedom give **p = 0.034**, which is below 0.05 — same conclusion, same threshold: a t of exactly 2.262 gives exactly p = 0.05.

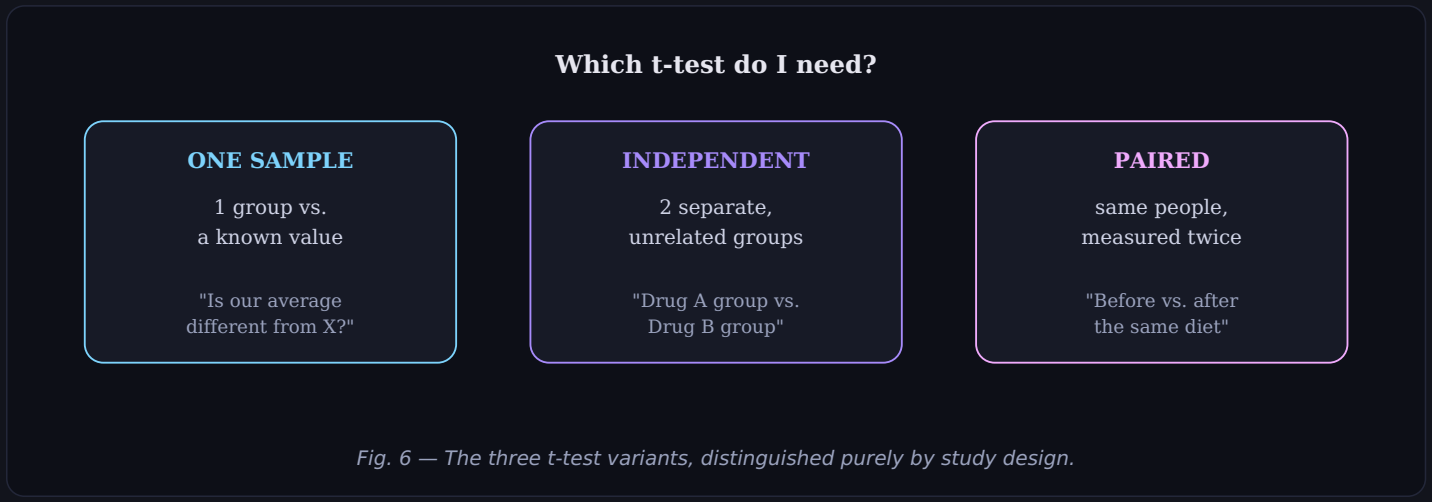


Fig. 6 — The three t-test variants, distinguished purely by study design.

One-Way ANOVA

The t-test only compares **two** groups. The moment you have **three or more** groups — fertilizers A, B, and C, say — you need **Analysis of Variance (ANOVA)**, which is a direct extension of the t-test.

HYPOTHESES

H_0: $\mu_1 = \mu_2 = \ldots = \mu_k$ (all group means are equal)

H_1: at least one μ is different

Assumptions

1. **Level of measurement:** independent variable (the grouping factor) is nominal; dependent variable is metric.
2. **Independence:** one group's measurements don't influence another's.
3. **Normality** within each group (tested with Shapiro-Wilk / Q-Q plot). Matters less as sample size grows. If violated → use the **Kruskal-Wallis test** instead.
4. **Homogeneity of variance** across groups (Levene's test). If violated → use **Welch's ANOVA**.

The Core Idea: Between-Group vs. Within-Group Variance

ANOVA compares two kinds of "spread": how much the group *means* differ from each other (between-group variance) versus how much individuals within each group differ from their own group's mean (within-group variance).

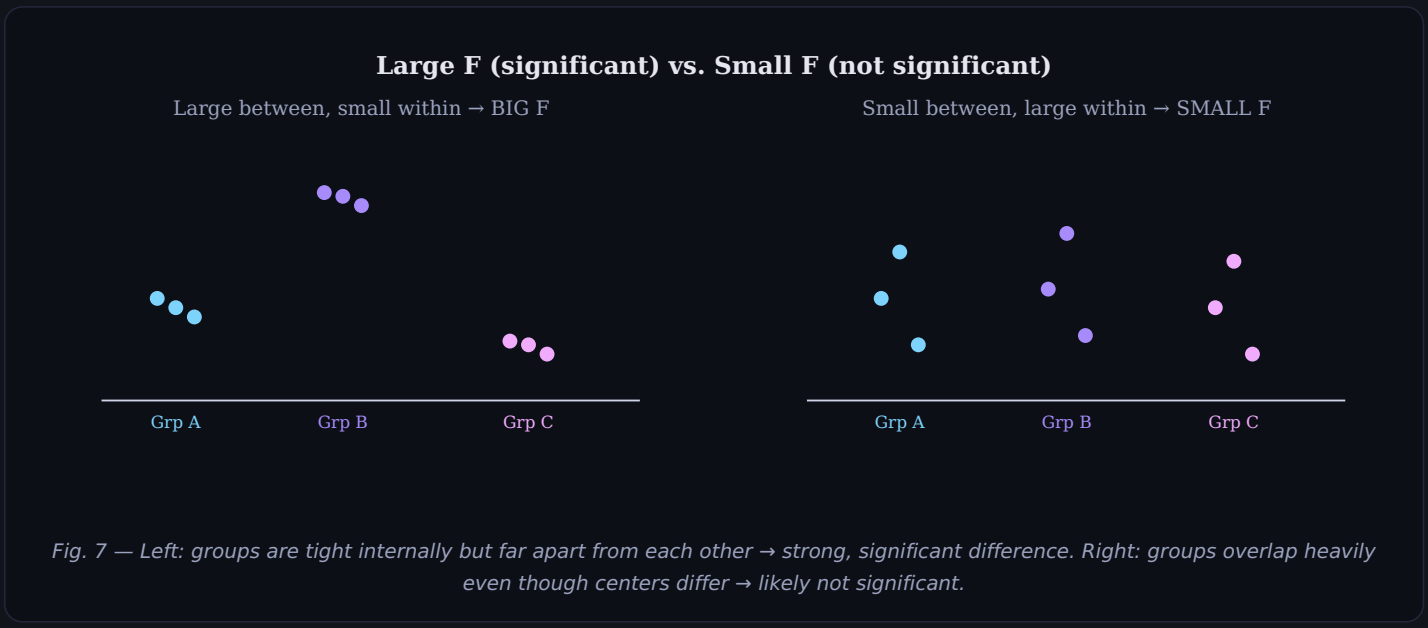


Fig. 7 — Left: groups are tight internally but far apart from each other → strong, significant difference. Right: groups overlap heavily even though centers differ → likely not significant.

The Formulas

F-STATISTIC

$$F = \frac{\text{Mean Square Between (MSB)}}{\text{Mean Square Within (MSW)}} = \frac{\text{variance between the groups}}{\text{variance within the groups}}$$

- **Sum of Squares Between (SSB)** = $\Sigma\, n_k(\bar{x}_k - \bar{x})^2$ — each group's mean, compared to the grand mean, weighted by group size.
- **Sum of Squares Within (SSW)** = $\Sigma(x_i - \bar{x}_k)^2$ — each individual value, compared to its own group's mean.
- **MSB** = SSB ÷ (k − 1), where k = number of groups.
- **MSW** = SSW ÷ (N − k), where N = total observations.

WORKED EXAMPLE — 3 GROUPS, N=8 EACH (N=24)

Grand mean = 5.3, group means = 2.5, 8.0, 5.5. SSB works out to a value that, divided by df = 2, gives **MSB = 60.67**. SSW divided by df = 21 gives **MSW ≈ 1.71**. That yields **F ≈ 35.39**, and with (2, 21) degrees of freedom this corresponds to **p < 0.01** — we reject H_0: the three groups genuinely differ.

Post-Hoc Tests

WHY YOU NEED THEM

A significant ANOVA only tells you *some* group differs from some other group — not *which* pair. A **post-hoc test** (e.g. Bonferroni) runs pairwise comparisons afterward to pinpoint exactly where the difference lies. Sometimes all groups differ from each other; sometimes only one pair does.

Two-Way, Repeated-Measures & Mixed ANOVA

Two-Way ANOVA

Used when you have **two categorical factors** influencing one metric dependent variable — e.g. does **drug type** (A/B) *and* **gender** (M/F) affect blood-pressure reduction? Each factor's categories are called its **levels**; combining a 2-level factor with a 2-level factor gives 4 groups to compare.

A two-way ANOVA answers **three** questions at once, each with its own H_0/H_1 pair:

1. Does Factor A (drug type) have a main effect?
2. Does Factor B (gender) have a main effect?
3. **Interaction effect:** does the effect of one factor *depend on the level of the other*? (E.g. Drug A helps men but hurts women, while Drug B does the opposite — averaged out, neither drug looks like it matters, but the interaction is huge.)

SPLITTING THE TOTAL VARIANCE

Total SS = SS(Factor A) + SS(Factor B) + SS(Interaction) + SS(Error). Each piece, divided by its own degrees of freedom, gives a mean square; each mean square divided by the error mean square gives an F-value — one F (and one p-value) per row.

Assumptions: normality (per cell, via Q-Q plot), homogeneity of variance (Levene's), independence, and metric dependent variable — the same family of checks as one-way ANOVA, just applied to every factor combination.

Repeated-Measures ANOVA

This is the ANOVA answer to "what if the same subjects are measured at **three or more** time points or conditions?" — the natural extension of the paired t-test the same way one-way ANOVA extends the independent t-test.

EXAMPLE

Physical fitness of the same volunteers measured **before** a training program, **immediately after**, and **two months later**. Three dependent (linked) samples from the same people.

HYPOTHESES

H_0: no difference between the time points/conditions (each person has the same underlying value at all times, even if that value differs between people).
H_1: there is a difference.

EXTRA ASSUMPTION — SPHERICITY

The variances of the *differences* between every pair of conditions must be roughly equal. Checked with **Mauchly's test**: $p > 0.05$ → sphericity holds. If violated, apply a **Greenhouse-Geisser** or **Huynh-Feldt** correction to the degrees of freedom, or switch to the non-parametric **Friedman test**.

Calculation splits variance into **within-subject** variance (SS Treatment, from the differences between time-point means and the grand mean) and **residual** variance (SS Residual = SS Within-subjects − SS Treatment), then forms F = MS(Treatment) / MS(Residual) exactly as before.

Mixed-Model ANOVA

What if you want a between-subjects factor (like therapy type, where each patient gets only *one* therapy) **and** a within-subjects factor (like time, where the same patients are re-measured) in a single analysis? That's a **mixed-model ANOVA** — literally a two-way ANOVA where one factor is a repeated measure.

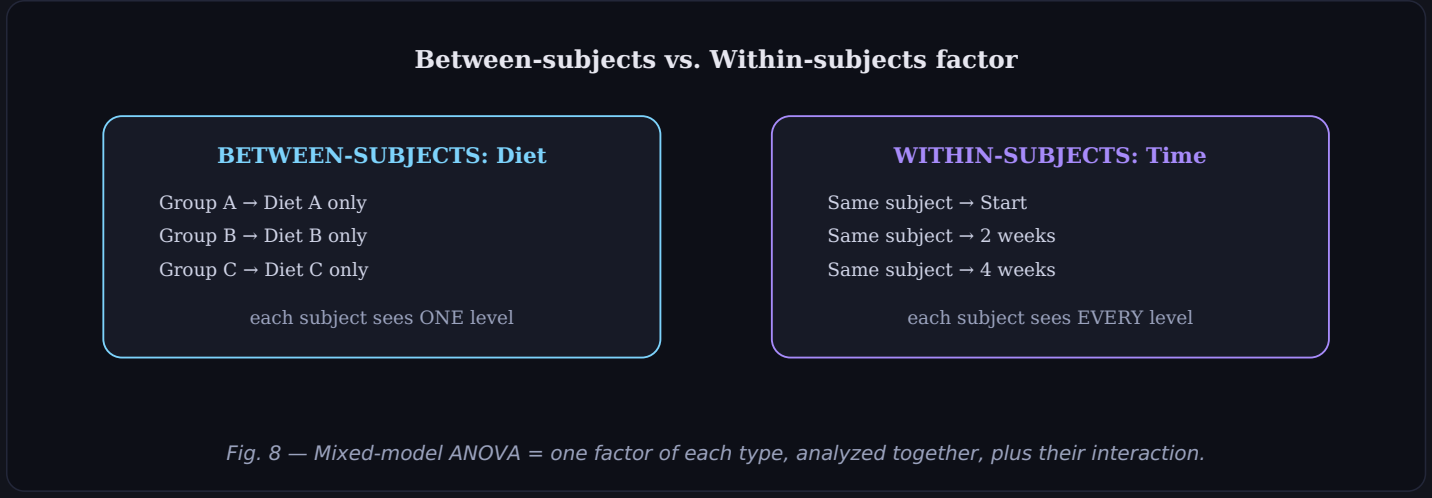


Fig. 8 — Mixed-model ANOVA = one factor of each type, analyzed together, plus their interaction.

THREE HYPOTHESES (SAME PATTERN AS TWO-WAY ANOVA)

1) The within-subject factor (time) has no effect. 2) The between-subjects factor (therapy) has no effect. 3) No interaction between them.

ASSUMPTIONS

Normality, homogeneity of variance (both between and within), **sphericity** for the within-subjects factor (Mauchly's test), independence of observations, and no significant outliers.

Non-Parametric Tests & Chi-Square

Every parametric test above has a non-parametric counterpart that swaps **raw values** for **ranks**. Ranking removes the need for normality — but at the cost of some statistical power. Use them only when the parametric assumptions genuinely fail.

Parametric test	Non-parametric counterpart	What differs
Independent t-test	**Mann-Whitney U test**	Compares rank sums instead of means
Paired t-test	**Wilcoxon signed-rank test**	Ranks the (unsigned) differences, then compares + vs. − rank sums
One-way ANOVA	**Kruskal-Wallis test**	Compares rank sums across 3+ independent groups
Repeated-measures ANOVA	**Friedman test**	Ranks within each subject/row, then compares rank sums across time points
Pearson correlation	Spearman's rho (see Part 5)	Correlates ranks instead of raw values

Mann-Whitney U Test

Sort *all* values from both groups together, smallest to largest, and assign ranks 1, 2, 3... Add up the ranks belonging to each group — those are the **rank sums (T)**.

FORMULA

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - T_1 \quad \text{(same for } U_2\text{; the smaller of } U_1, U_2 \text{ is the test statistic U)}$$

WORKED EXAMPLE — REACTION TIME, MEN VS. WOMEN

6 women, 5 men. Rank sum women (T_1) = 37, rank sum men (T_2) = 29. $U_1 = 6 \cdot 5 + (6 \cdot 7)/2 - 37 =$ **14**; U_2 works out to **16**. Test statistic U = min(14,16) = **14**. Expected U under $H_0 = n_1 n_2/2 = 15$. Standard error ≈ 5.48. z = (14 − 15)/5.48 ≈ **−0.18** → p ≈ 0.86, far above 0.05 — no significant difference.

Wilcoxon Signed-Rank Test

For paired data: compute each pair's difference, rank the *absolute* differences, then tag each rank with the sign (+/−) of its original difference. Sum the positive ranks (T^+) and negative ranks (T^-) separately; the test statistic **$W = \min(T^+, T^-)$**. Under H_0 the positive and negative rank sums should be roughly equal.

Kruskal-Wallis Test

Same idea as Mann-Whitney, extended to 3+ groups: rank everyone together, sum ranks per group, and compare those rank sums to what you'd expect if the groups didn't matter.

TEST STATISTIC H (≈ X^2 DISTRIBUTED)

$$H = \frac{12}{N(N+1)} \sum n_k(\bar{R}_k - \bar{R})^2$$

Compare H to a critical χ^2 value with (k − 1) degrees of freedom.

Friedman Test

Non-parametric repeated-measures: within each subject's row, rank the time points (not across subjects). Sum each time point's ranks down the column, then compare those column rank sums with a χ^2-based statistic — same structural logic as Kruskal-Wallis, just ranking within rows instead of across the whole sample.

The Chi-Square Test

All the tests so far need a **metric** dependent variable. What if *both* your variables are categorical — e.g. does gender influence preferred newspaper? That calls for the **chi-square (χ^2) test of independence**.

HYPOTHESES

H_0: there is no relationship between the two categorical variables (they're independent).
H_1: there is a relationship.

FORMULA

$$\chi^2 = \Sigma \frac{(O_i - E_i)^2}{E_i}$$

O = observed count in a cell · E = expected count if the variables were perfectly independent

ASSUMPTION

Every cell's **expected** frequency should be greater than 5. If many cells are small, the test becomes unreliable — consider merging categories or using Fisher's exact test.

Degrees of freedom = (rows − 1) × (columns − 1). Compare your χ^2 statistic to the critical χ^2 value for that many degrees of freedom at $\alpha = 0.05$ (or just read off the p-value).

WORKED EXAMPLE — GENDER × NEWSPAPER

4×2 contingency table → df = (4−1)(2−1) = 3. Calculated $\chi^2 = 0.504$. Critical χ^2 at α=0.05, df=3 is 7.815. Since $0.504 < 7.815$ (equivalently, $p = 0.918 > 0.05$), we **fail to reject H_0** — no evidence gender influences preferred newspaper in this sample.

Correlation Analysis

Correlation analysis measures the relationship between two variables — specifically, **how strong** it is and **which direction** it goes. Both answers live in a single number: the **correlation coefficient**, always between −1 and +1.

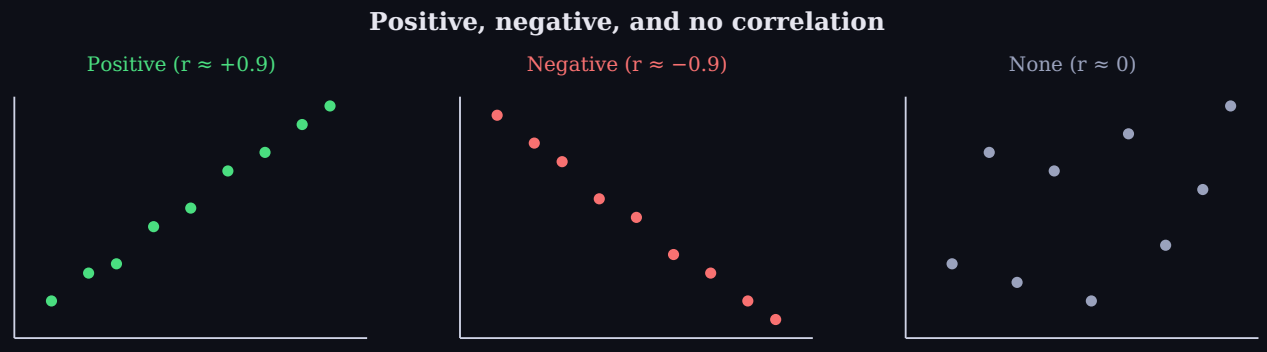


Fig. 9 — Positive: both variables rise together (height & shoe size). Negative: one rises as the other falls (price & sales). None: points scattered with no pattern.

|r|	Strength
0.0 – 0.1	No correlation
0.1 – 0.3	Weak
0.3 – 0.5	Moderate
0.5 – 0.7	Strong
0.7 – 1.0	Very strong

Pearson Correlation (r)

Measures the **linear** relationship between two **metric** variables.

FORMULA

$$r = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[\Sigma(x_i - \bar{x})^2 \cdot \Sigma(y_i - \bar{y})^2]}}$$

Intuition: subtract the mean from every x and every y, then multiply the pairs. Two positive deviations multiplied together give a positive number; two negatives also give a positive number (a point below-average on both variables still supports a *positive* correlation). A positive times a negative gives a negative number. If most points fall into the "both high" or "both low" quadrants, the sum — and hence r — comes out positive.

TESTING SIGNIFICANCE

H_0: the population correlation is 0 (no linear relationship). Tested via a t-statistic built from r and n; requires **both variables to be normally distributed** for the resulting p-value to be trustworthy.

Spearman's Rank Correlation (ρ / r_s)

The non-parametric counterpart of Pearson: instead of raw values, it correlates the **ranks**. Rank both variables, then simply compute Pearson correlation on the ranks. No normality required, and it works for ordinal data too.

SHORTCUT FORMULA (NO TIED RANKS)

$$r_s = 1 - \frac{6\,\Sigma d_i^2}{n(n^2 - 1)}$$

where d_i = the difference between the two rankings for each observation

Kendall's Tau (τ)

Also non-parametric and rank-based, but built differently — it counts **concordant vs. discordant pairs**. Preferred over Spearman when the sample is small and has many tied ranks.

FORMULA

$$\tau = \frac{C - D}{C + D}$$

C = concordant pairs (both rankings agree on order) · D = discordant pairs (they disagree)

WORKED EXAMPLE — TWO DOCTORS RANKING 6 PATIENTS

Comparing rank orders gives 11 concordant pairs and 4 discordant pairs. $\tau = (11 - 4)/(11 + 4) =$ **0.47** — a moderate positive agreement between the two doctors' rankings.

Point-Biserial Correlation (r_pb)

A special case of Pearson correlation for when one variable is **dichotomous** (two categories, e.g. pass/fail coded 1/0) and the other is **metric** (e.g. study hours).

NEAT FACT

Testing whether r_pb is significant gives you the *exact same p-value* as running an independent-samples t-test on the same two variables. They're mathematically two views of the same question.

Correlation vs. Causation

THE CLASSIC TRAP

Ice cream sales correlate strongly with sunburns. Ice cream doesn't cause sunburn, and sunburn doesn't cause ice cream cravings — a hidden third variable, **hot sunny weather**, drives both. A correlation alone never tells you which variable (if either) is the cause.

To claim **causality**, you need a significant correlation *plus* at least one of:

1. **Chronological order** — the cause clearly happened before the effect.
2. **A controlled experiment** — you deliberately manipulated one variable and observed the other change.
3. **A well-founded theory** explaining the mechanism.

CAUTIONARY TALE

Statistician Darryl Huff noted a *negative* correlation between head-lice counts and body temperature on an island. Islanders concluded lice cure fever. The true causal arrow ran the other way: high fever drives lice away. Correlation told them nothing about direction — only an experiment revealed the real mechanism.

Simple Linear Regression

Regression goes one step further than correlation: instead of just measuring how strongly two variables move together, it builds an equation that lets you **predict** one variable (the **dependent variable**, y) from another (the **independent variable**, x).

TWO USES OF REGRESSION

1) **Explanation** — measure how much influence x has on y (research-oriented).
2) **Prediction** — forecast y for a new case based on its x (application-oriented), e.g. predicting hospital stay length or which product a customer will buy.

The Regression Equation

FORMULA

$$\hat{y} = a + bx$$

a = intercept (predicted y when x = 0) · b = slope (change in y per one-unit increase in x)

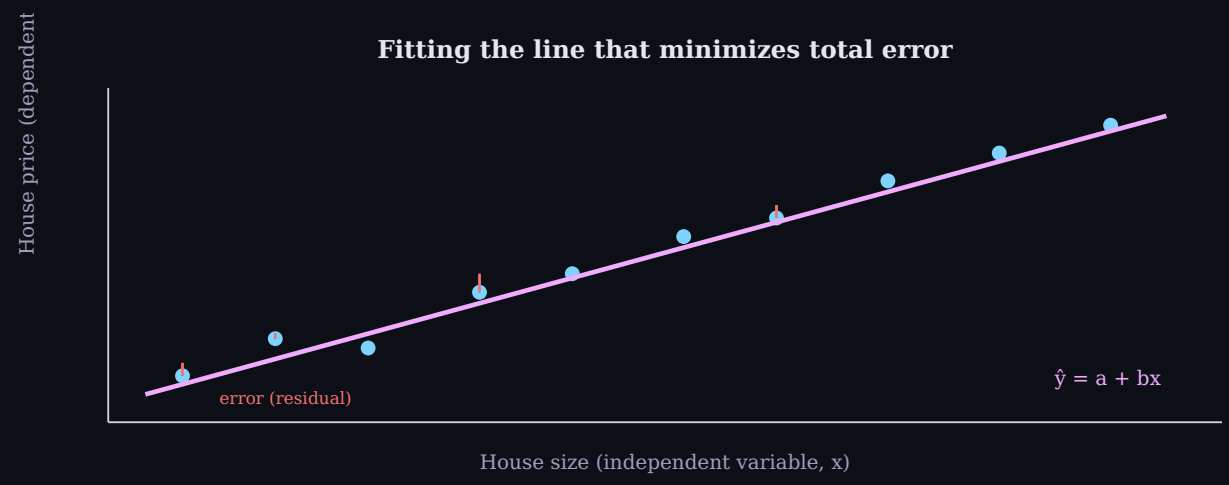


Fig. 10 — The regression line is the one that minimizes the total (squared) distance between the line and every actual data point — the red segments are the errors, or residuals.

CALCULATING B AND A

$$b = r \cdot \frac{s_y}{s_x} \qquad a = \bar{y} - b \cdot \bar{x}$$

r = correlation coefficient · s_y, s_x = standard deviations of y and x

WORKED EXAMPLE — HOUSE PRICE FROM HOUSE SIZE

r = 0.92 between size and price. Slope b = 18.35 (each extra square foot adds $18.35 to predicted price). Intercept a = 6,919.44. Equation: **$\hat{y}$ = 6919.44 + 18.35x**. So a 2,000 sq. ft. house is predicted at 6919.44 + 18.35×2000 ≈ **$43,619**.

Is the Relationship Significant?

The regression output includes a p-value testing H_0: "no relationship between x and y" (i.e., the true slope is 0). $p < 0.05$ → reject H_0 → the relationship is statistically significant.

Assumptions of Linear Regression

Assumption	What it means	How to check
Linearity	A straight line genuinely fits the data	Scatter plot of x vs. y
Independence of errors	One point's error doesn't affect another's	Durbin-Watson test
Homoscedasticity	Error spread is constant across all x values	Residuals plot (predicted vs. residuals)
Normally distributed errors	Residuals follow a normal distribution	Q-Q plot of residuals

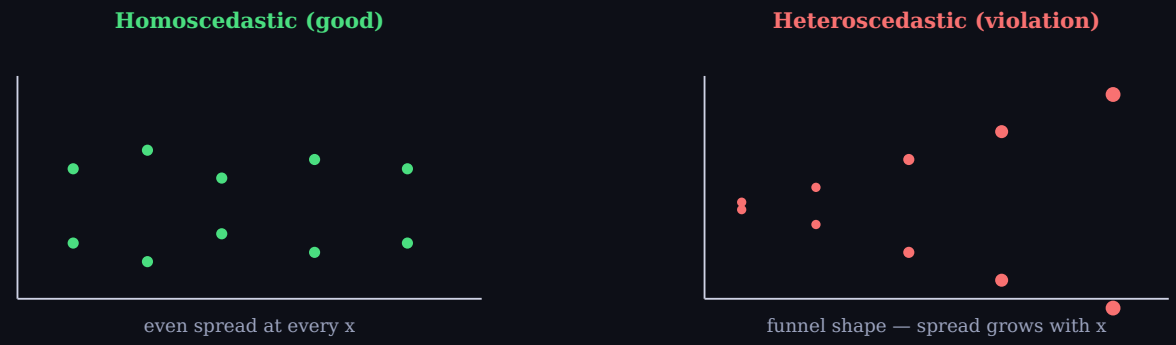


Fig. 11 — Homoscedasticity means the "fuzziness" around the regression line stays constant. A funnel shape (heteroscedasticity) means predictions get less reliable at higher x — a violated assumption.

Multiple Linear Regression

Real outcomes rarely depend on just one variable. **Multiple linear regression** extends simple linear regression to two or more independent (predictor) variables at once.

FORMULA

$$\hat{y} = a + b_1x_1 + b_2x_2 + \ldots + b_kx_k$$

Each b_k is interpreted "holding all other variables constant": e.g. b(age) = 0.26 means *for each extra year of age, blood pressure rises by 0.26 units, assuming weight and cholesterol stay fixed.*

Standardized vs. Unstandardized Coefficients

WHY STANDARDIZE?

Age (years) and cholesterol (mg/dL) live on wildly different scales, so their raw (unstandardized) coefficients can't be compared to see which predictor matters more. **Standardized coefficients** rescale every variable to the same units first, making their relative influence directly comparable.

Model Fit: R and R^2

Statistic	Meaning
R (multiple correlation)	Correlation between the actual y values and the model's predicted ŷ values. Higher = better fit.
R^2 (coefficient of determination)	Proportion of variance in y explained by the model. $R^2 = 0.52$ → the predictors explain 52% of the variation in the outcome.
Adjusted R^2	R^2 penalized for the number of predictors, so adding useless variables doesn't inflate it artificially.
Standard error of the estimate	Average distance between actual and predicted values, in the outcome's own units.

The Fifth Assumption: No Multicollinearity

THE PROBLEM

If two predictors are themselves highly correlated (e.g. house size and number of rooms), the model can't cleanly separate their individual effects on the outcome — the coefficients become unstable and hard to interpret, even though predictions can still be fine.

Detection: for each predictor, run a mini-regression predicting it from all the *other* predictors. A high R^2 there means that variable is redundant.

TOLERANCE & VIF

$$\text{Tolerance} = 1 - R^2 \qquad \text{VIF} = \frac{1}{\text{Tolerance}}$$

Tolerance < 0.1, or VIF > 10 → multicollinearity is a real concern

Fixes: drop one of the redundant predictors, or combine them into a single composite variable.

Categorical Predictors: Dummy Variables

Regression coefficients only make sense for numeric input, so a categorical predictor needs to be recoded first.

- **Two categories** (e.g. gender): code one as 0 (the **reference category**) and the other as 1. The coefficient then represents the difference between the two groups.
- **Three or more categories** (e.g. vehicle type: sedan / sports car / family van): create **k − 1 dummy variables** (here, 2), each a 0/1 flag for one category. The category with all flags at 0 is the reference. You never need a dummy for every category — if you know it's not a sedan and not a sports car, it must be the family van, so a third dummy would be redundant (and would itself cause multicollinearity).

Encoding "vehicle type" (3 categories) as 2 dummy variables

vehicle_type	is_sports_car	is_family_van
sedan	0	0
sports car	1	0
family van	0	1

Fig. 12 — Sedan is the reference category (both dummies = 0); the model's coefficients for "is_sports_car" and "is_family_van" then represent each group's difference from sedan.

Logistic Regression, Odds & Odds Ratios

Linear regression predicts a **metric** outcome. What if the outcome is **binary** — diseased/not diseased, spam/not spam, will-buy/won't-buy? A straight line would happily predict values like 1.4 or −0.3, which make no sense as probabilities. **Logistic regression** fixes this by squeezing the output into the range (0, 1) using the **logistic (sigmoid) function**.

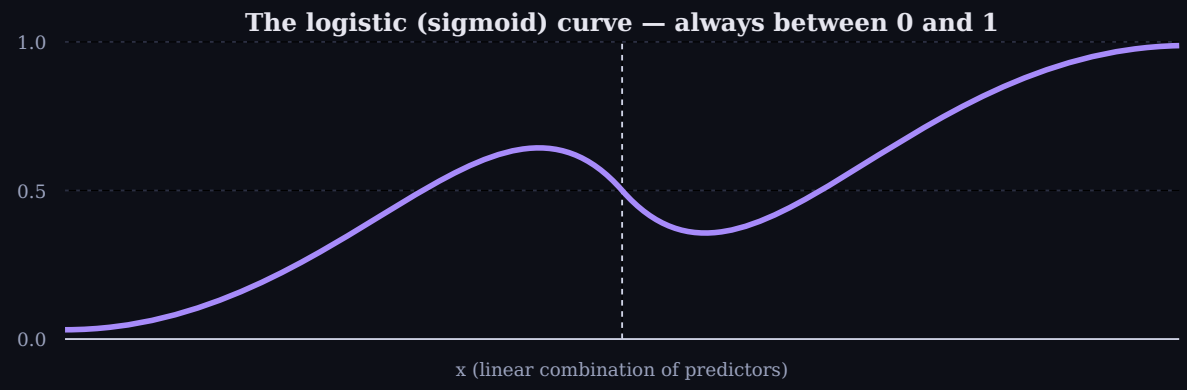


Fig. 13 — No matter how extreme x gets, the logistic function output stays trapped between 0 and 1 — exactly what a probability needs.

FORMULA

$$P(y=1) = \frac{1}{1 + e^{-(a + b_1x_1 + b_2x_2 + \ldots)}}$$

Coefficients (a, b_1, b_2…) are found via **maximum likelihood estimation** — a numerical method (not a closed-form formula like linear regression) that finds the coefficients making the observed data most probable.

EXAMPLE

Predicting whether a person has a certain disease from age, gender, and smoking status. For a 55-year-old male non-smoker, plugging the fitted coefficients into the formula might give **P(diseased) = 0.36**, i.e. 36%.

Reading the Output

- **Classification table** — how many cases the model correctly/incorrectly assigned, using a threshold (default 50%: probability > 0.5 → classified as the "1" outcome).
- **Chi-square test (model significance)** — compares the full model to a model with no predictors; a significant result means the predictors, together, genuinely help.
- **Pseudo-R^2** — logistic regression has no single agreed-upon R^2; common versions are Cox & Snell, Nagelkerke, and McFadden.
- **Per-predictor p-values** — same interpretation as always: $p < 0.05$ means that predictor's effect is unlikely to be zero.

Odds and Odds Ratios

ODDS (NOT THE SAME AS PROBABILITY!)

$$\text{Odds} = \frac{P(\text{event})}{P(\text{no event})}$$

If P(success) = 0.7, odds = 0.7 / 0.3 = **2.33** — success is 2.33× more likely than failure.

ODDS RATIO (OR)

Compares the odds of an event between two groups: OR = Odds(group A) ÷ Odds(group B).

- OR > 1 → the event is more likely in group A
- OR < 1 → the event is less likely in group A
- OR = 1 → no difference between the groups

WORKED EXAMPLE — MEDICATION AND DISEASE RISK

Group A (medication): P(diseased) = 0.6 → odds = 0.6/0.4 = 1.5. Group B (no medication): P(diseased) = 0.8 → odds = 0.8/0.2 = 4. OR = 1.5 / 4 = **0.38** — the odds of disease in the medicated group are 0.38× the odds in the unmedicated group (i.e. the medication is associated with lower odds of disease).

SHORTCUT FOR READING REGRESSION OUTPUT

In the results table, the odds ratio for any predictor is simply **e (raised to that predictor's coefficient)**. A coefficient of 0.04 for age gives $e^{0.04} \approx$ **1.04** — each extra year of age multiplies the odds of the outcome by 1.04.

K-Means Cluster Analysis

Every method so far tested a hypothesis you already had in mind. **Cluster analysis** flips the script: it looks for hidden groupings in data with **no predefined labels**. K-means is the simplest and most common clustering algorithm.

The Five Steps

1. **Choose K** — decide the number of clusters in advance (the "K" in K-means).
2. **Place K centroids randomly** among the data.
3. **Assign** every point to its *nearest* centroid.
4. **Recompute** each centroid as the mean position of all points currently assigned to it.
5. **Repeat steps 3–4** until assignments stop changing.



Fig. 14 — K-means repeatedly reassigns points and recentres the centroids until the clusters stop changing.

Choosing K: The Elbow Method

Adding more clusters always shrinks the total distance between points and their centroid — even a meaningless extra cluster helps a little. The **elbow method** plots total within-cluster distance against K, and you pick the point where the curve stops dropping steeply and "bends" — the point of diminishing returns.

CAVEAT

K-means starts from *random* initial centroids, so results can vary between runs. In practice, the algorithm is run many times from different random starting points, and the best (lowest-distance) solution is kept.

Confidence Intervals

A sample mean is only an **estimate** of the true population mean — different samples give different means. A **confidence interval (CI)** is a range built around that estimate meant to capture the true value.

THE INTERPRETATION ALMOST EVERYONE GETS WRONG

It is **tempting** — but technically incorrect (in the classical/frequentist framework) — to say "there is a 95% chance the true mean lies in this interval." Once an interval is calculated, the true population mean either **is** or **isn't** inside it; there's no probability left to assign to a fixed number.

THE CORRECT INTERPRETATION

If you repeated the sampling process many times and built a 95% CI each time, **95% of those intervals** would contain the true population value (and 5% wouldn't). The 95% confidence describes the reliability of the *method*, not any single interval you've already computed.

20 samples, 20 confidence intervals — most capture the true mean



Fig. 15 — Each horizontal line is one sample's 95% CI. Blue intervals capture the true mean (dashed line); red ones miss it. Across many repeated samples, about 95% turn out blue — that's what "95% confidence" really refers to.

Calculating a Confidence Interval for the Mean

FORMULA (NORMALLY DISTRIBUTED DATA)

$$\mathrm{CI} = \bar{x} \pm z \cdot \frac{s}{\sqrt{n}}$$

$\bar{x}$ = sample mean · z = critical value for the confidence level · s = std. dev. · n = sample size

For a **95% confidence level**, the critical value is **z = 1.96** (read from the standard normal distribution table). So the interval is $\bar{x} \pm 1.96 \times (s/\sqrt{n})$ — the "±" gives you the lower and upper bound.

Frequentist vs. Bayesian: A Quick Note

FREQUENTIST (WHAT THIS TUTORIAL COVERS)

Treats the true population parameter as **fixed but unknown**. Probability statements apply to the sampling procedure, not to the fixed parameter itself — hence the careful CI wording above.

BAYESIAN

Treats the parameter itself as a random variable with its own probability distribution, updated using a **prior distribution** plus the observed data. Its analogue to the CI is the **credible interval**, which *can* be interpreted as "there's a 95% probability the parameter lies in this range" — but the result depends

on the (somewhat subjective) choice of prior.